ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 1 · PRACTICE D

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET first paper. Section A (30): multiple choice, true or false, matching Python terms and Scratch block categories · Section B (70): hardware and software, algorithms, Git branches, variables, binary numbers, a Scratch maths game to study, an ATM flowchart and a Python program with ten errors to fix.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Start each section on a new page.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which ONE of the following is an output device?

A Keyboard B Monitor C Mouse D Scanner

1.2 1 024 bytes are equal to one …

A kilobyte. B megabyte. C gigabyte. D bit.

1.3 Which ONE of the following is an open-source operating system?

A Windows 10 B macOS C iOS D Linux

1.4 Which Python operator raises a number to a power?

A ^ B // C ** D %

1.5 What does print(10 // 3) display?

A 3 B 3.33 C 1 D 30

1.6 Which Scratch block makes the blocks inside it run over and over without stopping?

A wait B forever C say D glide

1.7 Which flowchart symbol is used for start and end?

A Diamond B Parallelogram
C Oval (terminator) D Rectangle

1.8 Which Git command uploads your commits to GitHub?

A git init B git status
C git add D git push

1.9 IDE stands for …

A Integrated Development Environment. B Internal Data Entry.
C Interactive Design Editor. D Input Device Emulator.

1.10 The decimal number 5 written in binary is …

A 100. B 101. C 110. D 111.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 A Python variable name may start with a digit.

2.2 A Python list is written inside square brackets.

2.3 Scratch sprites can be programmed to react when a key is pressed.

2.4 Python's input() function automatically returns a number.

2.5 The command git add prepares (stages) changes before they are committed.

(5)

[5]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Repeats code a fixed number of times	A	==
3.2	Changes a value into a decimal (floating-point) number	B	list
3.3	Takes a given item out of a list	C	function
3.4	Joins two strings together	D	for loop
3.5	Tests whether two values are equal	E	print()
3.6	A named block of code that can be used again	F	remove()
3.7	Data type for whole numbers	G	continue
3.8	Displays a value on the screen	H	float()
3.9	An ordered collection of items that can be changed	I	int
3.10	Skips to the next repetition of a loop	J	+

(10)

[10]

QUESTION 4

Name the Scratch block category (for example Control) in which each of the following blocks is found.

4.1 move 10 steps

4.2 say Hello! for 2 seconds

4.3 play sound Meow until done

4.4 when this sprite clicked

4.5 set score to 0

(5)

[5]

TOTAL SECTION A: [30]

SECTION B

QUESTION 5

5.1 Answer the following questions:

5.1.1 What does RAM stand for? (1)

5.1.2 Which device connects a home or office network to the internet? (1)

5.1.3 Name ONE storage device that has no moving parts. (1)

5.1.4 What is the function of the power supply unit (PSU) in a computer? (1)

5.2 Explain the difference between hardware and software, and give TWO examples of each. (4)

5.3 Explain what an algorithm is, and give TWO characteristics of a good algorithm. (4)

5.4 Explain what a branch is in Git, and why programmers in a team use branches. (4)

5.5 Explain what a variable is, and the difference between a variable and a constant. (4)

5.6 Convert the binary number 1101101 to decimal. Show your working. (5)

[25]

QUESTION 6

Study the Scratch program below and answer the questions that follow.

```
when green flag clicked
set [lives] to 3
set [score] to 0
repeat until < (lives) = 0 >
    set [a] to (pick random 1 to 10)
    set [b] to (pick random 1 to 10)
    ask (join (a) (join [ + ] (b))) and wait
    if < (answer) = ((a) + (b)) > then
        change [score] by 1
        say [Correct!] for 1 seconds
    else
        change [lives] by -1
        say (join [Wrong! Lives left: ] (lives)) for 1 seconds
```

```
end repeat
say (join [Game over! Score: ] (score)) for 3 seconds
```

6.1 Name the FOUR variables in the program and say what each one stores. (4)

6.2 When does the repeat loop stop? Explain. (4)

6.3 Explain what happens to the variables when the user answers correctly and when the user answers wrongly. (4)

6.4 The sums asked are 3 + 4, then 5 + 5, then 2 + 9, then 6 + 1, then 8 + 8. The user answers 7, 9, 10, 7 and 15. Write down the final message that the sprite says, and show how you got it. (4)

6.5 Suggest how the program can be changed to give the player one extra life after every 5 correct answers. (4)

[20]

QUESTION 7

You are asked to design the logic of a simple cash machine (ATM). A customer inserts a card and enters a PIN. If the PIN is wrong, the machine shows "Incorrect PIN" and returns the card. If the PIN is correct, the customer enters an amount. If the amount is more than the balance, the machine shows "Insufficient funds"; otherwise it gives the cash and updates the balance.

Draw a flowchart that represents the following process:

7.1 The process starts when a customer inserts a card. (2)

7.2 The customer enters a PIN. (2)

7.3 If the PIN is wrong, "Incorrect PIN" is shown, the card is returned and the process ends. (3)

7.4 If the PIN is correct, the customer enters the amount. (2)

7.5 If the amount is more than the balance, "Insufficient funds" is shown; otherwise the cash is given and the balance is reduced by the amount. (5)

7.6 The process ends. (1)

[15]

QUESTION 8

The Python program below should read the temperature for each of FIVE days, work out the average and say whether the week was hot, cold or mild. The program contains TEN errors (syntax and logic). Write down the line number and describe each error (for example: Line 3 – variable not declared) in the ANSWER BOOK.

```
1  temps = []
2  for day in range(1, 5):
3      t = input("Temperature for day " + day + ": ")
4      temps.add(t)
5  total = "0"
6  for t in temp:
7      total = total + t
8  average = total / 5
9  if average > 30
```

```
10      print("Hot week")
11  elif average < 15:
12  print("Cold week")
13  else:
14      print("Mild week"
15  print("Average: " + average)
```

(10)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER